

TELCO AUTONOMOUS RESOLVE-X

Autonomous Closed-Loop Telecom Operations Agent

Detect → Correlate → Reason → Plan → Act → Validate → Re-plan

Multi-agent AI prototype for autonomous telecom incident resolution. Safe simulator; no production network access.

The Problem

Telecom incidents are detected by many signals but still require fragmented human coordination.

1. Signal overload

Alarms, KPIs, service logs and symptoms arrive from different systems.

2. Slow diagnosis

Engineers correlate symptoms, identify root cause and search runbooks manually.

3. Manual action

Remediation requires hand-offs across network and service teams.

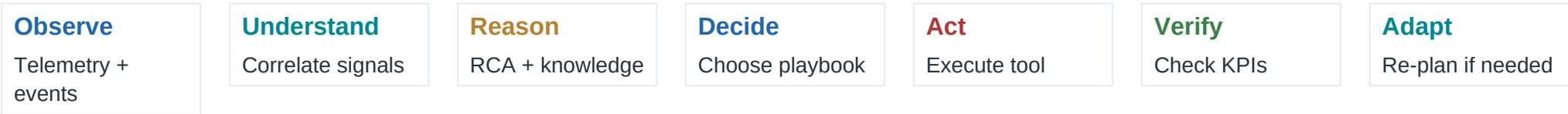
4. Weak feedback loop

Recovery is not always automatically re-validated after the first action.

Resolve-X turns the incident lifecycle into an autonomous decision loop with evidence, action tools, validation and audit memory.

Our Solution

An agentic telecom operations system that works toward an outcome, not just an answer.



Key differentiator: the system can continue operating after the first action. Validation feeds the next decision automatically.

Multi-Agent Collaboration

Specialized agents each own a distinct decision responsibility.

Signal Monitoring Agent

Detect anomaly and establish incident state

RCA + Knowledge Agent

Infer root cause and retrieve supporting evidence

Autonomous Decision Agent

Select next action from approved playbooks

Incident Correlation Agent

Join KPI symptoms into one failure pattern

Customer Impact Agent

Estimate users / sessions at risk

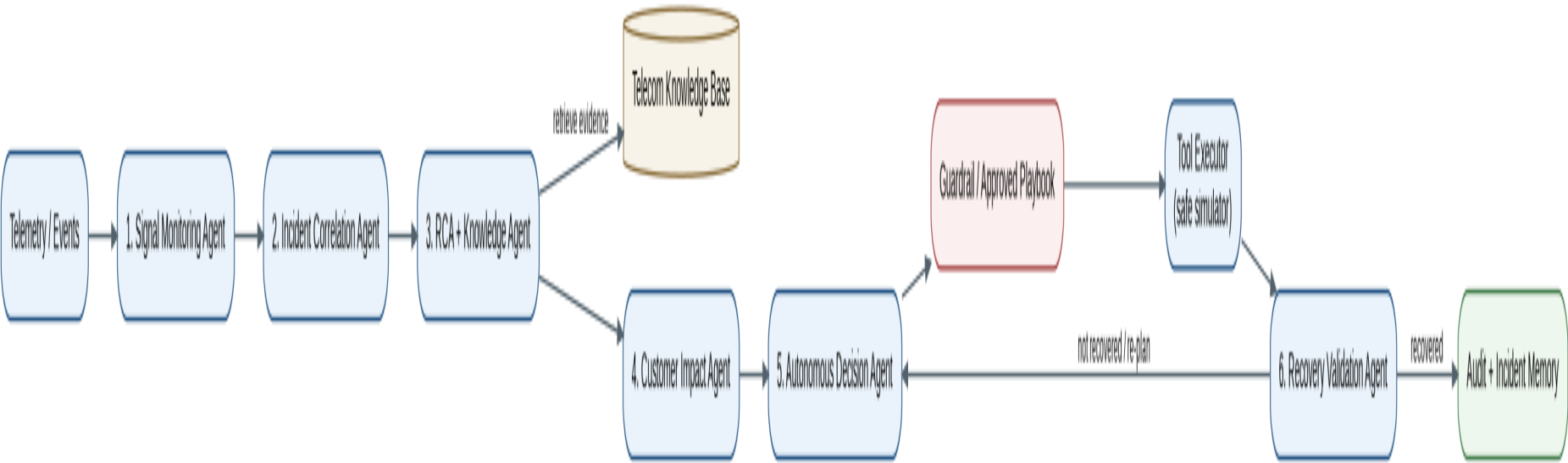
Recovery Validation Agent

Verify KPI recovery and trigger re-plan

Shared state + audit trace connect all agents through the orchestrator.

Technical Architecture

Closed-loop orchestration with a governed action layer.



Working Prototype - Four Telecom Scenarios

The simulator provides repeatable incidents so judges can observe autonomous actions and validation.

Scenario	Failure	Autonomous flow	Outcome
INC-1001	Backhaul congestion	Reroute → validate → fallback recovery	RESOLVED
INC-1002	Packet-core saturation	Scale capacity → validate	RESOLVED
INC-1003	OTP gateway instability	Restart gateway → validate	RESOLVED
INC-1004	Cell overload	Rebalance load → validate	RESOLVED

Smoke test: all four scenarios reached RESOLVED in the simulator.

Autonomy Without Human Intervention

Autonomy is constrained by explicit action guardrails rather than an unrestricted agent.

Autonomous decisions

Choose remediation from an allow-list; execute the next action when validation fails; maintain incident state; close only after KPI checks pass.

Safety guardrails

Approved playbook only; no production credentials; simulator tools; deterministic validation thresholds; full audit trail.

Production path

Replace simulator with governed OSS/BSS APIs; add policy/RBAC; add observability and rollback; route high-risk actions to enterprise approval policies.

Hackathon demo: zero human intervention in the simulated recovery loop.

Prototype Stack

Designed for a fast, reproducible hackathon build.

UI

Streamlit dashboard

Knowledge

Local telecom knowledge base + evidence retrieval

Tools

Simulated remediation APIs / action playbooks

Deployment

Local / container-ready pattern

Orchestration

Python agent coordinator + shared state

AI

Optional OpenAI model for narrative summarisation

Memory

JSON incident audit memory

Hackathon Evaluation Alignment

The attached ET AI guidance emphasises AI effectiveness, agentic capability, technical execution, impact, prototype quality and responsible AI.

Significance & relevance

Telecom operations / customer-impact problem

Effective use of AI

RCA reasoning + evidence retrieval + agent planning

Agentic autonomy

Reason → plan → act → recover from failure → close

Prototype quality

Interactive UI + 4 deterministic demo scenarios

Innovation & originality

Closed-loop multi-agent self-healing workflow

Technical complexity

Stateful orchestration, tools, validation and re-planning

Business impact

MTTR reduction potential and fewer manual hand-offs

Responsible AI

Allow-listed tools, audit trail, no production access

Business Impact & Enterprise Potential

From incident detection to autonomous recovery in one operational loop.

Operational value

Reduce manual triage and cross-team handoffs. Accelerate time from detection to corrective action. Create a consistent, auditable recovery process.

Customer value

Faster restoration. Proactive customer communication. Lower risk of prolonged service degradation.

Scale path

OSS/BSS integrations. Streaming telemetry. Network controllers. Policy + RBAC + rollback. Cloud agent runtime.

Measure next: MTTR, autonomous-resolution rate, first-action success rate, false remediation rate, customer-impact minutes avoided.

2-4 Minute Demo Storyboard

Use INC-1001 for the strongest judge flow.

Time	Scene	What to say/show
0:00-0:20	Problem	Telecom incidents create fragmented alarms and manual coordination.
0:20-0:45	Trigger	Select INC-1001 and show KPI anomalies.
0:45-1:20	Reason	Show monitoring → correlation → RCA + evidence.
1:20-1:55	Act	Decision Agent selects reroute; no human approval.
1:55-2:30	Validate	Validation checks KPIs; if needed, fallback action runs.
2:30-3:15	Impact	Show resolved state, audit trace and enterprise path.

Submission Package

Ready to be placed into the user GitHub repository and demonstrated live.

1. Working prototype

GitHub-ready Python + Streamlit. Four telecom scenarios. Unit-tested autonomous flow. README + demo instructions.

2. Pitch deck

This deck in PowerPoint and PDF. Architecture. Innovation + impact. Evaluation alignment.

3. Demo video

Record the Streamlit UI using the supplied 2-4 minute script. Show INC-1001 end-to-end with the agent trace visible.

Target repository: github.com/garv4179/Agentic-Ai-telecom-domain-ET-AI-Hackathon

Positioning note: the attached detailed problem-statement PDF names Banking, Cybersecurity, Retail and Utilities as the four detailed problems; telecom is not listed in that PDF. This deck therefore positions telecom as your selected domain concept while aligning to the general agentic evaluation framework.